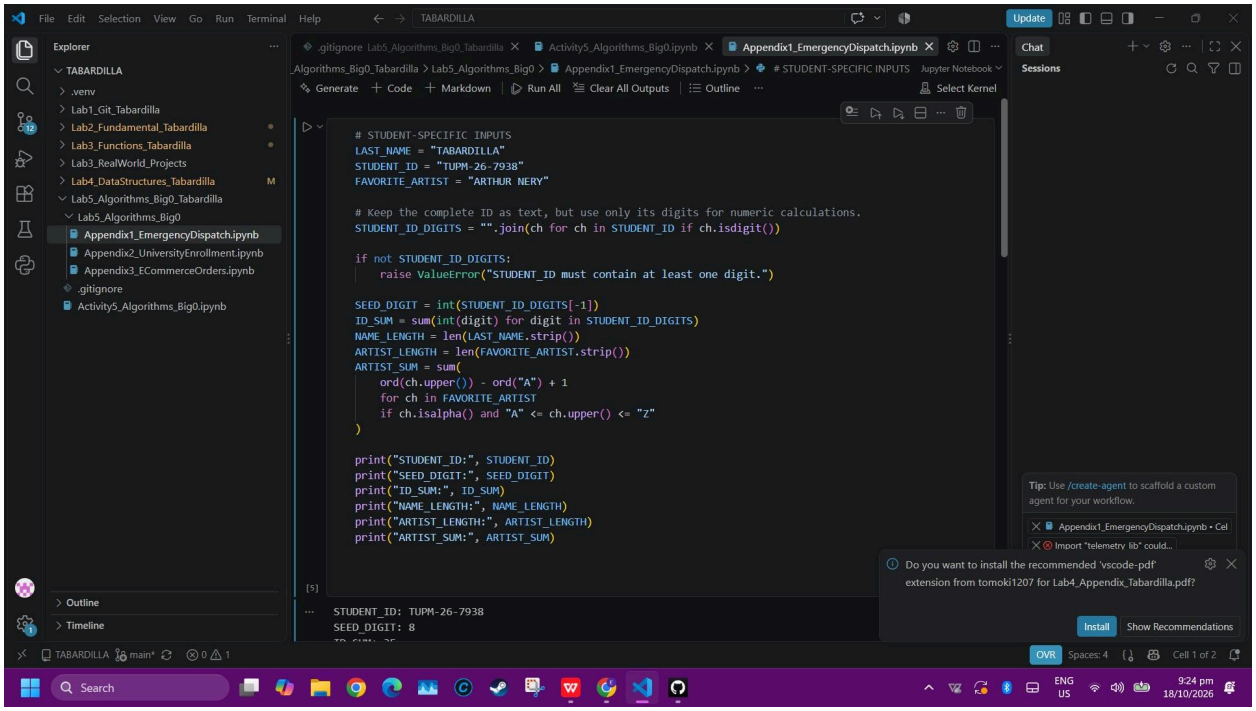
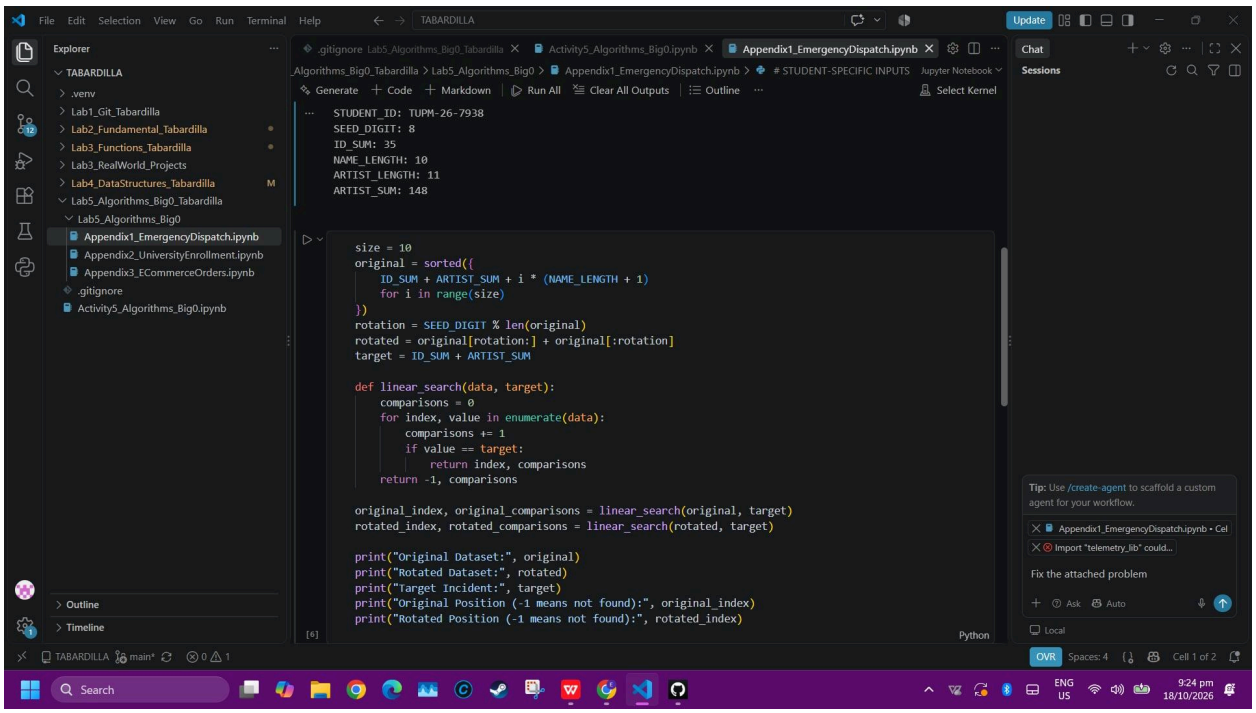
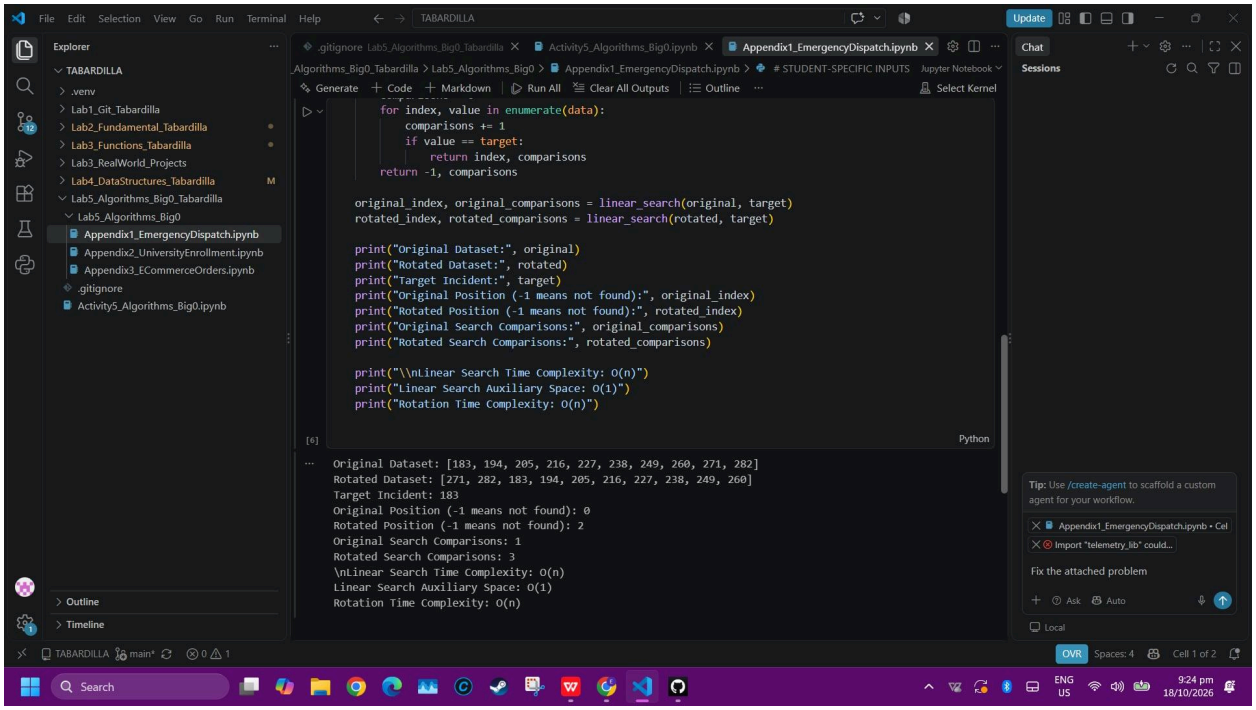
Erika S. Tabardilla
BSECE - 1C

Activity 5

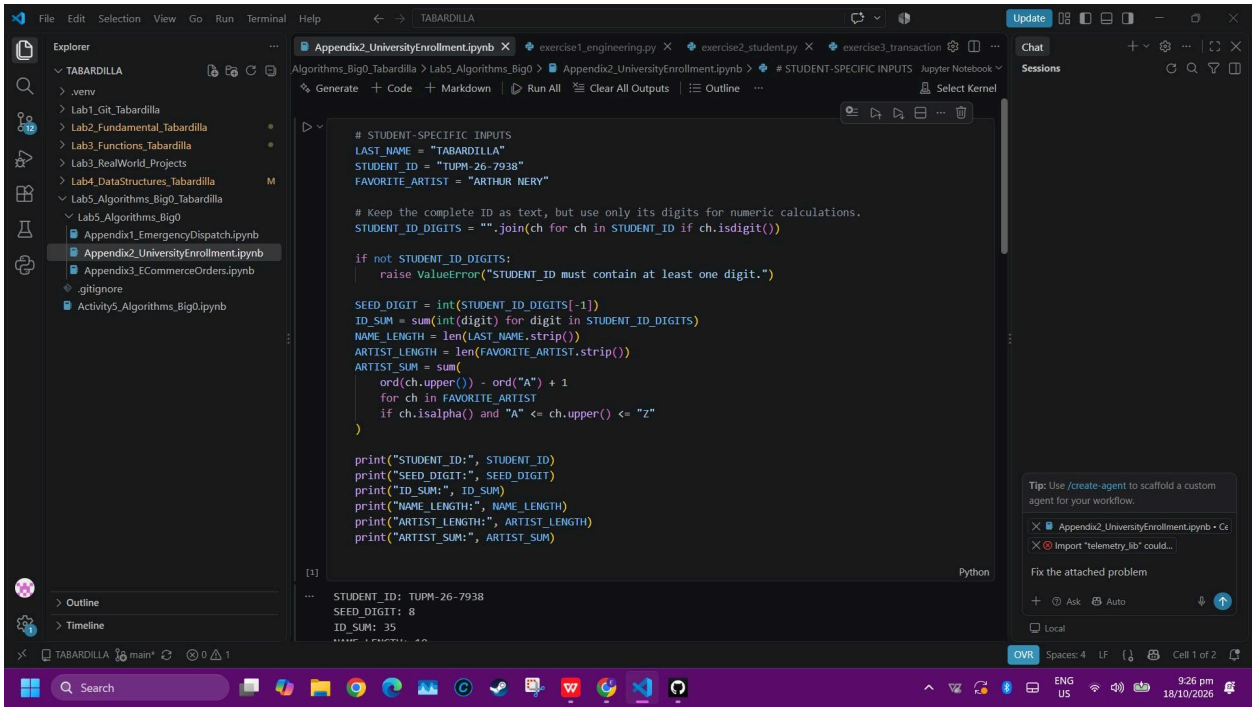
APPENDIX 1:

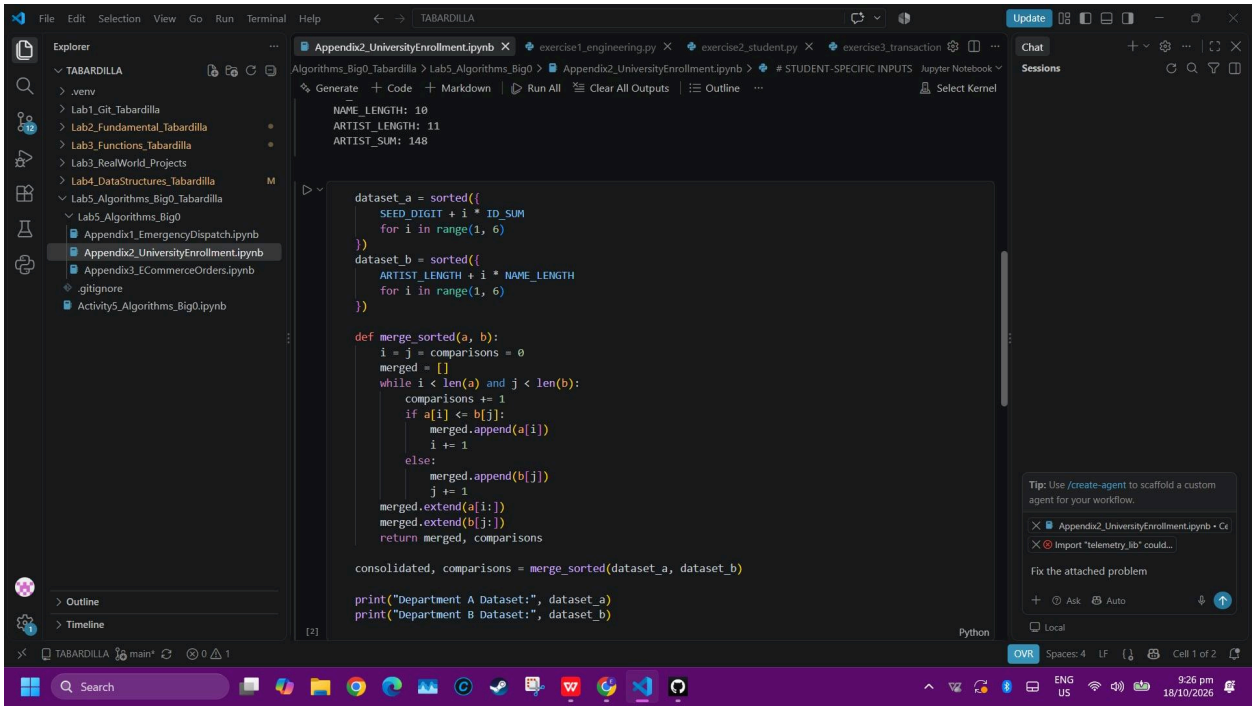


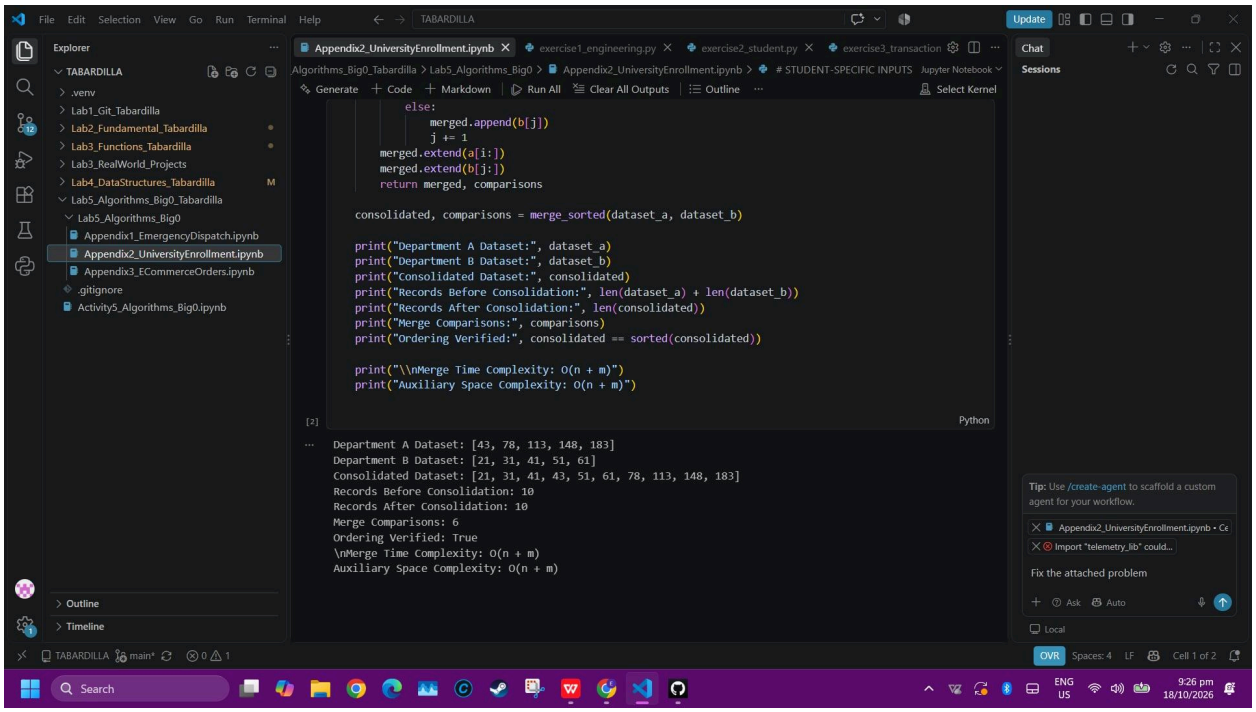



for index, value in enumerate(data):
comparisons += 1
if value == target:
return index, comparisons
return -1, comparisons
original_index, original_comparisons = linear_search(original, target)
rotated_index, rotated_comparisons = linear_search(rotated, target)
print("Original Dataset:", original)
print("Rotated Dataset:", rotated)
print("Target Incident:", target)
print("Original Position (-1 means not found):", original_index)
print("Rotated Position (-1 means not found):", rotated_index)
print("Original Search Comparisons:", original_comparisons)
print("Rotated Search Comparisons:", rotated_comparisons)
print("\\nLinear Search Time Complexity: O(n)")
print("Linear Search Auxiliary Space: O(1)")
print("Rotation Time Complexity: O(n)")
Original Dataset: [183, 194, 205, 216, 227, 238, 249, 260, 271, 282]
Rotated Dataset: [271, 282, 183, 194, 205, 216, 227, 238, 249, 260]
Target Incident: 183
Original Position (-1 means not found): 0
Rotated Position (-1 means not found): 2
Original Search Comparisons: 1
Rotated Search Comparisons: 3
\nLinear Search Time Complexity: O(n)
Linear Search Auxiliary Space: O(1)
Rotation Time Complexity: O(n)

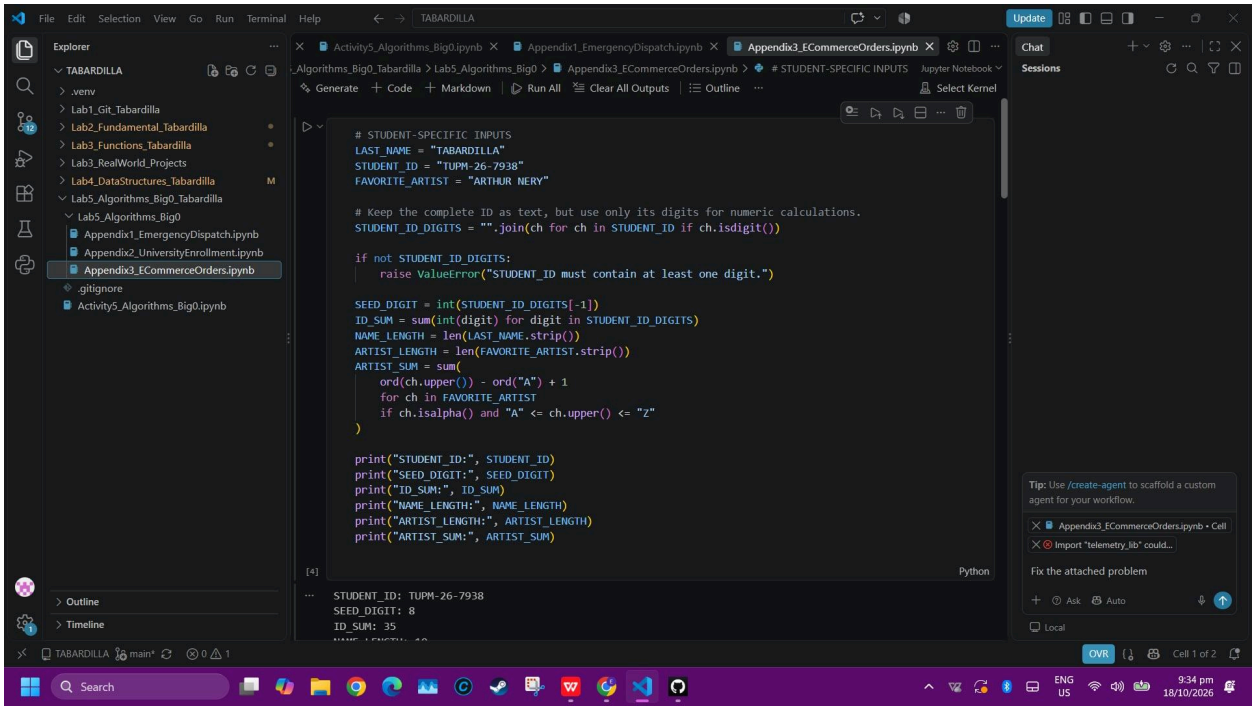
APPENDIX 2:

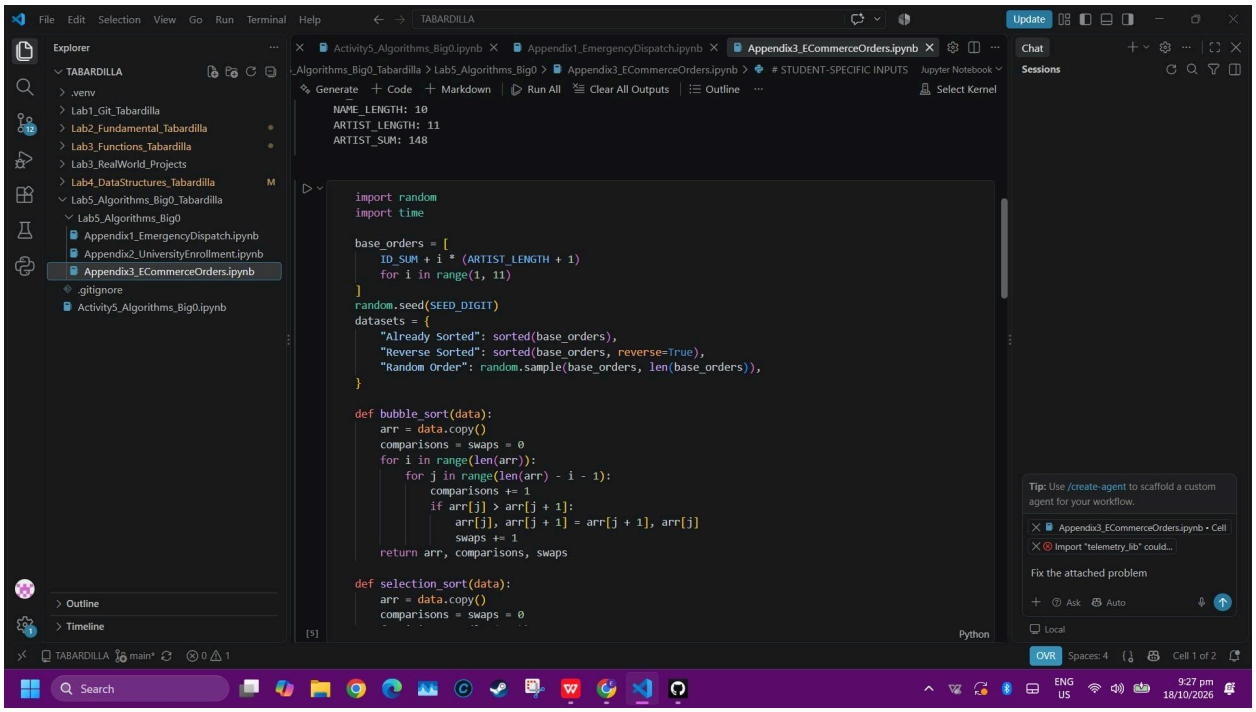

STUDENT-SPECIFIC INPUTS
LAST_NAME = "TABARDILLA"
STUDENT_ID = "TUPM-26-7938"
FAVORITE_ARTIST = "ARTHUR NERY"
Keep the complete ID as text, but use only its digits for numeric calculations.
STUDENT_ID_DIGITS = "".join(ch for ch in STUDENT_ID if ch.isdigit())
if not STUDENT_ID_DIGITS:
raise ValueError("STUDENT_ID must contain at least one digit.")
SEED_DIGIT = int(STUDENT_ID_DIGITS[-1])
ID_SUM = sum(int(digit) for digit in STUDENT_ID_DIGITS)
NAME_LENGTH = len(LAST_NAME.strip())
ARTIST_LENGTH = len(FAVORITE_ARTIST.strip())
ARTIST_SUM = sum(
ord(ch.upper()) - ord("A") + 1
for ch in FAVORITE_ARTIST
if ch.isalpha() and "A" <= ch.upper() <= "Z"
)
print("STUDENT_ID:", STUDENT_ID)
print("SEED_DIGIT:", SEED_DIGIT)
print("ID_SUM:", ID_SUM)
print("NAME_LENGTH:", NAME_LENGTH)
print("ARTIST_LENGTH:", ARTIST_LENGTH)
print("ARTIST_SUM:", ARTIST_SUM)
STUDENT_ID: TUPM-26-7938
SEED_DIGIT: 8
ID_SUM: 35

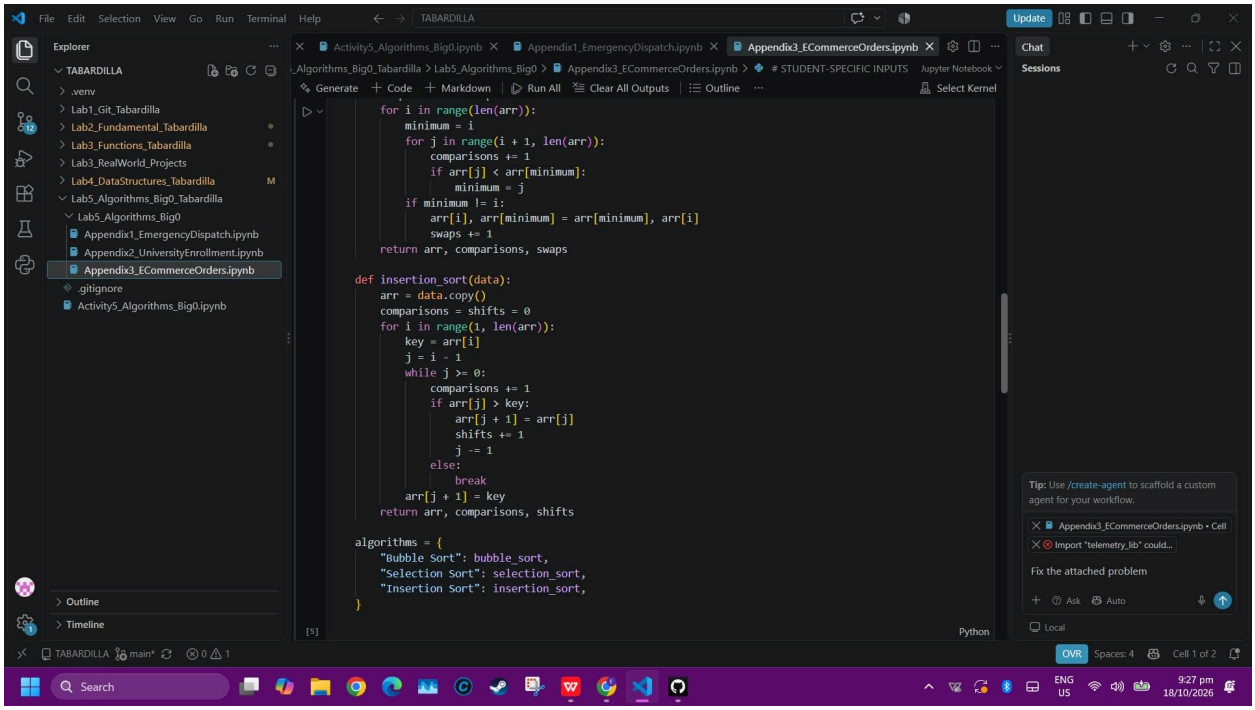

NAME_LENGTH: 10
ARTIST_LENGTH: 11
ARTIST_SUM: 148
dataset_a = sorted({
SEED_DIGIT + i * ID_SUM
for i in range(1, 6)
})
dataset_b = sorted({
ARTIST_LENGTH + i * NAME_LENGTH
for i in range(1, 6)
})
def merge_sorted(a, b):
i = j = comparisons = 0
merged = []
while i < len(a) and j < len(b):
comparisons += 1
if a[i] <= b[j]:
merged.append(a[i])
i += 1
else:
merged.append(b[j])
j += 1
merged.extend(a[i:])
merged.extend(b[j:])
return merged, comparisons
consolidated, comparisons = merge_sorted(dataset_a, dataset_b)
print("Department A Dataset:", dataset_a)
print("Department B Dataset:", dataset_b)
Python

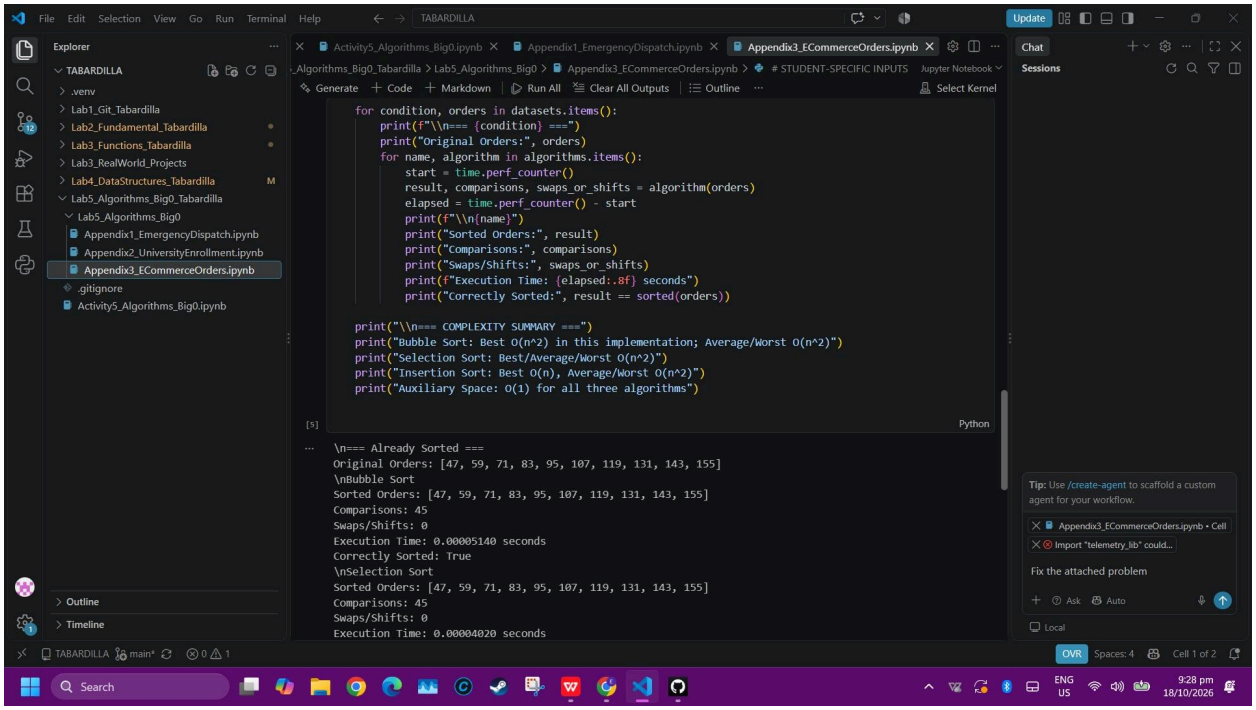

else:
merged.append(b[j])
j += 1
merged.extend(a[i:])
merged.extend(b[j:])
return merged, comparisons
consolidated, comparisons = merge_sorted(dataset_a, dataset_b)
print("Department A Dataset:", dataset_a)
print("Department B Dataset:", dataset_b)
print("Consolidated Dataset:", consolidated)
print("Records Before Consolidation:", len(dataset_a) + len(dataset_b))
print("Records After Consolidation:", len(consolidated))
print("Merge Comparisons:", comparisons)
print("Ordering Verified:", consolidated == sorted(consolidated))
print("\\nMerge Time Complexity: O(n + m)")
print("Auxiliary Space Complexity: O(n + m)")
Python
Department A Dataset: [43, 78, 113, 148, 183]
Department B Dataset: [21, 31, 41, 51, 61]
Consolidated Dataset: [21, 31, 41, 43, 51, 61, 78, 113, 148, 183]
Records Before Consolidation: 10
Records After Consolidation: 10
Merge Comparisons: 6
Ordering Verified: True
\nMerge Time Complexity: O(n + m)
Auxiliary Space Complexity: O(n + m)

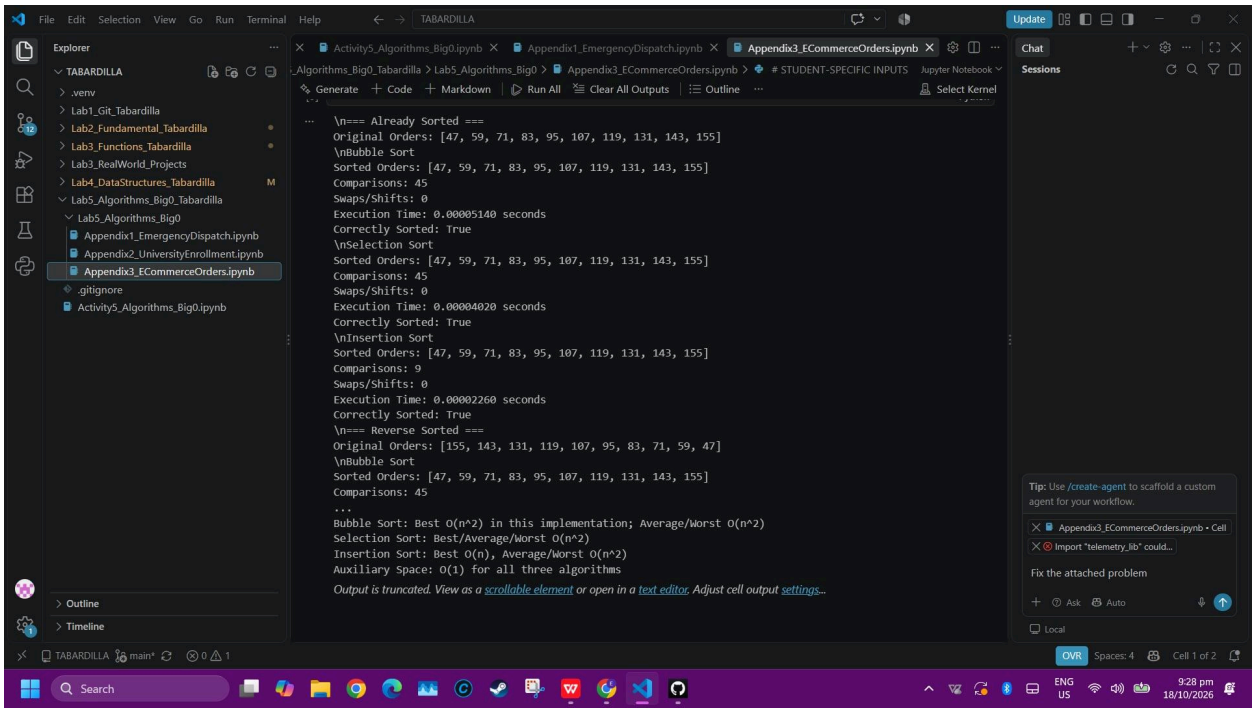
APPENDIX 3:


File Edit Selection View Go Run Terminal Help
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Lab5_Algorithms_Big0
Appendix1_EmergencyDispatch.ipynb
Appendix2_UniversityEnrollment.ipynb
Appendix3_ECommerceOrders.ipynb
.gitignore
Activity5_Algorithms_Big0.ipynb
Activity5_Algorithms_Big0.ipynb
Appendix1_EmergencyDispatch.ipynb
Appendix3_ECommerceOrders.ipynb
Algorithms_Big0_Tabardilla > Lab5_Algorithms_Big0 > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Clear All Outputs Outline
Select Kernel
STUDENT-SPECIFIC INPUTS
LAST_NAME = "TABARDILLA"
STUDENT_ID = "TUPM-26-7938"
FAVORITE_ARTIST = "ARTHUR NERY"
Keep the complete ID as text, but use only its digits for numeric calculations.
STUDENT_ID_DIGITS = "".join(ch for ch in STUDENT_ID if ch.isdigit())
if not STUDENT_ID_DIGITS:
raise ValueError("STUDENT_ID must contain at least one digit.")
SEED_DIGIT = int(STUDENT_ID_DIGITS[-1])
ID_SUM = sum(int(digit) for digit in STUDENT_ID_DIGITS)
NAME_LENGTH = len(LAST_NAME.strip())
ARTIST_LENGTH = len(FAVORITE_ARTIST.strip())
ARTIST_SUM = sum(
ord(ch.upper()) - ord("A") + 1
for ch in FAVORITE_ARTIST
if ch.isalpha() and "A" <= ch.upper() <= "Z"
)
print("STUDENT_ID:", STUDENT_ID)
print("SEED_DIGIT:", SEED_DIGIT)
print("ID_SUM:", ID_SUM)
print("NAME_LENGTH:", NAME_LENGTH)
print("ARTIST_LENGTH:", ARTIST_LENGTH)
print("ARTIST_SUM:", ARTIST_SUM)
[4]
Python
STUDENT_ID: TUPM-26-7938
SEED_DIGIT: 8
ID_SUM: 35
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Appendix3_ECommerceOrders.ipynb • Cell
Import "telemetry_lib" could...
Fix the attached problem
Ask Auto
Local
Outline
Timeline
TABARDILLA main* 0 1
OVR
Cell 1 of 2
Search
ENG US
9:34 pm 18/10/2026


File Edit Selection View Go Run Terminal Help
TABARDILLA
Update
Explorer
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Activity5_Algorithms_Big0.ipynb
Activity5_Algorithms_Big0.ipynb
Appendix1_EmergencyDispatch.ipynb
Appendix3_ECommerceOrders.ipynb
Algorithms_Big0_Tabardilla > Lab5_Algorithms_Big0 > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Clear All Outputs Outline
Select Kernel
NAME_LENGTH: 10
ARTIST_LENGTH: 11
ARTIST_SUM: 148
import random
import time
base_orders = [
ID_SUM + i * (ARTIST_LENGTH + 1)
for i in range(1, 11)
]
random.seed(SEED_DIGIT)
datasets = {
"Already Sorted": sorted(base_orders),
"Reverse Sorted": sorted(base_orders, reverse=True),
"Random Order": random.sample(base_orders, len(base_orders)),
}
def bubble_sort(data):
arr = data.copy()
comparisons = swaps = 0
for i in range(len(arr)):
for j in range(len(arr) - i - 1):
comparisons += 1
if arr[j] > arr[j + 1]:
arr[j], arr[j + 1] = arr[j + 1], arr[j]
swaps += 1
return arr, comparisons, swaps
def selection_sort(data):
arr = data.copy()
comparisons = swaps = 0
[5]
Python
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Appendix3_ECommerceOrders.ipynb • Cell
Import "telemetry_lib" could...
Fix the attached problem
Ask Auto
Local
Outline
Timeline
TABARDILLA main* 0 1
OVR
Spaces: 4
Cell 1 of 2
Search
ENG US
9:27 pm 18/10/2026


File Edit Selection View Go Run Terminal Help
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Outline
Timeline
Activity5_Algorithms_Big0.ipynb
Appendix1_EmergencyDispatch.ipynb
Appendix3_ECommerceOrders.ipynb
Generate
Code
Markdown
Run All
Clear All Outputs
Outline
Select Kernel
for i in range(len(arr)):
minimum = i
for j in range(i + 1, len(arr)):
comparisons += 1
if arr[j] < arr[minimum]:
minimum = j
if minimum != i:
arr[i], arr[minimum] = arr[minimum], arr[i]
swaps += 1
return arr, comparisons, swaps
def insertion_sort(data):
arr = data.copy()
comparisons = shifts = 0
for i in range(1, len(arr)):
key = arr[i]
j = i - 1
while j >= 0:
comparisons += 1
if arr[j] > key:
arr[j + 1] = arr[j]
shifts += 1
j -= 1
else:
break
arr[j + 1] = key
return arr, comparisons, shifts
algorithms = {
"Bubble Sort": bubble_sort,
"Selection Sort": selection_sort,
"Insertion Sort": insertion_sort,
}
[5]
Python
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Appendix3_ECommerceOrders.ipynb • Cell
Import "telemetry_lib" could...
Fix the attached problem
Ask
Auto
Local
TABARDILLA main*
0 1
OVR
Spaces: 4
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Search
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18/10/2026


File Edit Selection View Go Run Terminal Help
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Lab5_Algorithms_Big0_Tabardilla > Lab5_Algorithms_Big0 > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Clear All Outputs Outline
Select Kernel
for condition, orders in datasets.items():
print(f"\\n=== {condition} ===")
print("Original Orders:", orders)
for name, algorithm in algorithms.items():
start = time.perf_counter()
result, comparisons, swaps_or_shifts = algorithm(orders)
elapsed = time.perf_counter() - start
print(f"\\n{name}")
print("Sorted Orders:", result)
print("Comparisons:", comparisons)
print("Swaps/Shifts:", swaps_or_shifts)
print(f"Execution Time: {elapsed:.8f} seconds")
print("Correctly Sorted:", result == sorted(orders))
print("\\n=== COMPLEXITY SUMMARY ===")
print("Bubble Sort: Best O(n^2) in this implementation; Average/Worst O(n^2)")
print("Selection Sort: Best/Average/Worst O(n^2)")
print("Insertion Sort: Best O(n), Average/Worst O(n^2)")
print("Auxiliary Space: O(1) for all three algorithms")
[5]
Python
\n=== Already Sorted ===
Original Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
\nBubble Sort
Sorted Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00005140 seconds
Correctly Sorted: True
\nSelection Sort
Sorted Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00004020 seconds
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Appendix3_ECommerceOrders.ipynb • Cell
Import "telemetry_lib" could...
Fix the attached problem
Ask Auto
Local
Outline
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OVR Spaces: 4 Cell 1 of 2
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ENG US
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File Edit Selection View Go Run Terminal Help
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Lab5_Algorithms_Big0_Tabardilla > Lab5_Algorithms_Big0 > Appendix3_ECommerceOrders.ipynb > # STUDENT-SPECIFIC INPUTS
Jupyter Notebook
Generate Code Markdown Run All Clear All Outputs Outline
Select Kernel
\n=== Already Sorted ===
Original Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
\nBubble Sort
Sorted Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00005140 seconds
Correctly Sorted: True
\nSelection Sort
Sorted Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
Comparisons: 45
Swaps/Shifts: 0
Execution Time: 0.00004020 seconds
Correctly Sorted: True
\nInsertion Sort
Sorted Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
Comparisons: 9
Swaps/Shifts: 0
Execution Time: 0.00002260 seconds
Correctly Sorted: True
\n=== Reverse Sorted ===
Original Orders: [155, 143, 131, 119, 107, 95, 83, 71, 59, 47]
\nBubble Sort
Sorted Orders: [47, 59, 71, 83, 95, 107, 119, 131, 143, 155]
Comparisons: 45
...
Bubble Sort: Best O(n^2) in this implementation; Average/Worst O(n^2)
Selection Sort: Best/Average/Worst O(n^2)
Insertion Sort: Best O(n), Average/Worst O(n^2)
Auxiliary Space: O(1) for all three algorithms
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
Chat
Sessions
Tip: Use /create-agent to scaffold a custom agent for your workflow.
Appendix3_ECommerceOrders.ipynb • Cell
Import "telemetry_lib" could...
Fix the attached problem
Ask Auto
Local
Outline
Timeline
TABARDILLA main* 0 1
OVR Spaces: 4 Cell 1 of 2
Search
ENG US
9:28 pm 18/10/2026